

Advance Publication

The Journal of Veterinary Medical Science

Accepted Date: 7 April 2026

J-STAGE Advance Published Date: 15 April 2026

©2026 The Japanese Society of Veterinary Science

Author manuscripts have been peer reviewed and accepted for publication but have not yet been edited.

Note

A case of oral squamous cell carcinoma treated by mandibulectomy in a spotted seal (*Phoca largha*)

Yuki KIMURA¹⁾, Keisuke OKAMOTO²⁾, Masatoshi TSUNOKAWA²⁾, Minoru OKAMOTO³⁾,

Kazuya MATSUDA⁴⁾, Tadashi SANO⁵⁾, Toshikazu SAKAI¹⁾,

¹⁾Department of Companion Animal Clinical Sciences, School of Veterinary Medicine, Rakuno Gakuen University, Midori-machi 582, Bunkyo-dai, Ebetsu City, Hokkaido 069-0836, Japan

²⁾Otaru Aquarium Co. Ltd., Syukutsu, Otaru City, Hokkaido 047-0047, Japan

³⁾Department of Veterinary Clinical Pathology, Rakuno Gakuen University, Midori-machi 582, Bunkyo-dai, Ebetsu City, Hokkaido 069-0836, Japan

⁴⁾Department of Veterinary Pathology, Rakuno Gakuen University, Midori-machi 582, Bunkyo-dai, Ebetsu City, Hokkaido 069-0836, Japan

⁵⁾ Department of Agriculture and Veterinary Medicine, Division of Clinical Veterinary Medicine, Obihiro University, Inada-cho, Obihiro 080-8555, Japan

Corresponding Author: Toshikazu Sakai

Department of Veterinary Medicine, School of Veterinary Medicine, Rakuno Gakuen University,

Midori-machi 582, Bunkyo-dai, Ebetsu city, Hokkaido 069-0836, Japan

Fax/Phone: +81-11-388-4871

E-mail: t-sakai@rakuno.ac.jp

Running Head: MANDIBLECTOMY IN A SPOTTED SEAL

1 ABSTRACT

2 Most cases of squamous cell carcinoma (SCC) in pinnipeds are diagnosed postmortem;
3 antemortem diagnosis and active therapeutic intervention are rarely reported, and prognosis is generally
4 considered poor. We report an antemortem diagnosis of mandibular oral SCC in a spotted seal (*Phoca*
5 *largha*) with progressive feeding difficulty due to tumor-associated pain that worsened despite
6 analgesics. Partial rostral mandibulectomy (left second incisor to right second premolar) with lip-tissue
7 reconstruction restored feeding by postoperative day 2. Despite tumor recurrence, the seal maintained
8 feeding function and good quality of life for approximately 18 months. This first reported
9 mandibulectomy in a pinniped suggests that surgery may be a valuable palliative option for oral SCC.

10 **KEY WORDS:** case report; mandibulectomy; pinniped; spotted seal; squamous cell carcinoma

Squamous cell carcinoma (SCC) in pinnipeds has been reported in various anatomical sites, including the tongue, gingiva, esophagus, lungs, vagina, and pancreas [2, 4, 6–8, 10, 12]. However, most of these cases were diagnosed postmortem, and only a few reports have described antemortem diagnosis [2, 4, 10, 12]. Furthermore, to the best of our knowledge, no previous reports have documented active therapeutic intervention for SCC in pinnipeds, and the prognosis has generally been considered poor.

An approximately 34-year-old female spotted seal (*Phoca largha*), weighing approximately 80 kg, kept at Otaru Aquarium (Otaru, Japan) was brought to us with mandibular erythema and swelling. The seal had a history of cataracts but was otherwise clinically unremarkable. Neither emaciation nor obesity was evident on physical examination. Periodontal disease was initially suspected and treated for 7 days with prednisolone (10 mg BID, Prednisolone 5-mg tablets “Mita”; KYORIN Rimedio Co., Ltd., Kanazawa, Japan) and clindamycin (600 mg BID, Dalacin Capsules 150 mg; Pfizer Japan Inc., Tokyo, Japan). Both medications were administered orally by placing the tablets and capsules into fish and delivering them through the gills. Although transient improvement was observed, the symptoms recurred within 1 month, accompanied by bleeding and progressive swelling. The seal subsequently developed inconsistent feeding behavior and eventually stopped eating entirely, although appetite partially improved after analgesic administration. Blood samples were collected from the plantar venous plexus under physical restraint on days 7, 63, and 78, with the day of onset defined as day 1. Hematologic and

serum biochemical examinations revealed no remarkable abnormalities on days 7 and 63; however, leukocytosis was observed on day 78 (white blood cell counts: 6,000 cells/ μ L on day 7, 9,600 cells/ μ L on day 63, and 14,000 cells/ μ L on day 78). Physical examination identified a prominent gingival mass around the right mandibular canine tooth that displaced the tooth and obstructed the view of the two right mandibular incisors. The lesion surface was ulcerated (Fig. 1a).

On day 80 after onset, computed tomography (CT) and punch biopsy were performed under general anesthesia; the animal weighed 98 kg at that time. Based on the animal's age, clinical condition, and disease course, the anesthetic risk was classified as American Society of Anesthesiologists (ASA) physical status III. As premedication, intramuscular injections were administered in the gluteal region while the animal was inside the cage, using a long 21-gauge needle. Medetomidine (20 μ g/kg, Dolbene; Kyoritsu Seiyaku Corp., Tokyo, Japan), midazolam (0.2 mg/kg, Midazolam; Sandoz K.K., Tokyo, Japan), and butorphanol (0.2 mg/kg, Vetorphale; Meiji Seika Pharma Co., Ltd., Tokyo, Japan) were mixed and given intramuscularly (IM). The animal was removed from the cage and transferred to the CT room. During transport and until endotracheal intubation, 100% oxygen was supplied via face mask. After intubation, anesthesia was maintained with sevoflurane (0.75–2%, Sevoflo; Zoetis Japan Inc., Tokyo, Japan) in 100% oxygen during CT scan and biopsy. An intravenous catheter (22-gauge) was placed in the plantar interdigital vein located between the first and second digits of the left hind flipper, through which lactated Ringer's solution was administered intravenously at a rate of 3.7 mL/kg/hr.

Anesthetic reversal was achieved using flumazenil (0.4 mg/kg IM, Anexate; Sandoz K.K.). During general anesthesia, a marked decrease in heart rate was observed; this was managed by intravenous administration (IV) of atipamezole (100 µg/kg, Atipame Injection; Kyoritsu Seiyaku Corp.) and reduction of the sevoflurane concentration. Apart from this event, general anesthesia was completed without any particular complications.

CT revealed a 25 × 30 × 45-mm mass with indistinct margins, osteolysis, and bone proliferation around the right mandibular canine root, along with mandibular enlargement (Fig. 1b, 1c). Minimal contrast enhancement was observed, and no enlargement of regional lymph nodes, including the mandibular lymph nodes, was detected, suggesting no detectable metastasis. Biopsy was performed using a punch instrument, and hemostasis was achieved with bipolar electrocautery. Histopathological examination of the biopsy specimen confirmed a diagnosis of SCC.

Following CT examination, tramadol (Tramal Injection; Nippon Shinyaku Co., Ltd., Kyoto, Japan) was administered IM for 4 days (2 mg/kg BID for the first 2 days, followed by 2.5 mg/kg BID for the remaining 2 days); however, the seal did not resume feeding. Therefore, starting 5 days after CT examination and continuing until the day before surgery, buprenorphine (Buprenorphine Injection 0.3 mg “Nissin”; NISSIN Pharmaceutical Co., Ltd., Tendo, Japan) was administered IM for 58 days to control pain and maintain pre-intervention feeding ability, with the dosage gradually increased from 0.006 mg/kg BID to 0.012 mg/kg BID as needed. Intramuscular injections were administered in the

gluteal region, with injection sites rotated as much as possible. After the 24th injection, an approximately 10-cm-diameter abscess was noted in the right gluteal region, which was considered likely to be an injection-related complication. The abscess was punctured with an 18-gauge needle and managed by repeated manual expression. No clinical signs suggestive of injection-associated pain, including inappetence, were observed.

On day 143 after onset, partial mandibulectomy was performed. The seal exhibited weight loss due to poor feeding, and body weight at the time of surgery was 89.5 kg. ASA physical status was classified as III, consistent with that assigned during CT examination and biopsy. The seal was manually restrained and premedicated with a mixture of medetomidine (10 µg/kg), midazolam (0.1 mg/kg), and butorphanol (0.1 mg/kg, Vetorphale; Meiji Seika Pharma Co., Ltd., Tokyo, Japan), administered IM (total volume: 4.5 mL). The doses of the three anesthetic agents were reduced to half of those used during the previous anesthesia because no nociceptive responses had been observed during the biopsy under the prior anesthetic protocol, and to minimize the risk of excessive decreases in heart rate. However, adequate sedation was not achieved. Therefore, an additional dose of medetomidine (10 µg/kg IM) was administered approximately 8 min later, and the same dose was administered again 21 min after the first injection. An intravenous catheter was placed in the plantar interdigital vein located between the first and second digits of the left hind flipper. At 43 min 46 s after the initial premedication, anesthesia was induced with alfaxalone (0.25 mg/kg IV, Alfaxan Multidose; Meiji Animal Health Co., Ltd., Tokyo,

Japan), and the animal was removed from the cage and transferred to the operating room. During transport and until endotracheal intubation, sevoflurane was administered via a face mask, after which endotracheal intubation was performed using a cuffed 51-Fr tube (inner diameter, 12 mm).

The primary surgical objective was palliative pain relief via marginal excision rather than radical resection. The labial and sublingual mucosa were incised several millimeters from the tumor margin, and the mandibular body was exposed using monopolar and bipolar electrocautery (Fig. 2a). The left mandible was transected between the incisor and canine using bone scissors, while the right mandible was sectioned between the second and third premolars using a high-speed surgical drill (Surgairtome TWO; Hall Surgical, CONMED Corp, Utica, NY, USA) and an osteotome (Fig. 2b). Hemorrhage from the inferior alveolar vessels was controlled with bipolar cautery, and bone wax was applied to the mandibular canal for hemostasis (Fig. 2c). The genioglossus, geniohyoid, and mylohyoid muscles were transected, and the rostral portion of the mandible was completely removed (Fig. 2d). Reconstruction was performed by layered closure of the lip skin, subcutaneous tissue, and oral and sublingual mucosa using 3-0 polydioxanone sutures (Ethicon, Johnson & Johnson, Somerville, NJ, USA), restoring the mandibular contour as closely as possible to its original shape (Fig. 3).

During general anesthesia for surgery, a decrease in heart rate similar to that observed during CT examination and biopsy was noted and managed by atipamezole (228 µg/kg IV). No other major complications occurred, and no reversal agents were administered to facilitate recovery from anesthesia

during this procedure. The total anesthetic time from premedication to extubation was 252 min, and the recovery time from the end of anesthesia to extubation was 13 min; spontaneous respiration resumed 1 min 18 s after discontinuation of sevoflurane.

Postoperative medication was administered buprenorphine (0.015 mg/kg IV, Repetan; Otsuka Pharmaceutical Co., Ltd., Tokyo, Japan) for analgesia. Histopathological examination of the resected mandible confirmed the absence of tumor cells at the surgical margins, with resection performed through the inflammatory zone (Fig. 4).

By postoperative day 2 (with the day of surgery defined as postoperative day 0), the seal resumed feeding, reached its prescribed food intake by day 11, and was observed capturing and swallowing fish with the rostral portion of the mouth. Buprenorphine was tapered and discontinued by day 19, and food intake was maintained thereafter. Clinically, local recurrence was first noted 68 days after surgery as a localized gingival mass with marked erythema and mucosal ulceration. Mild bleeding was observed from the ulcerated surface. However, the seal maintained good appetite and stable feeding for more than 1 year without analgesics. On postoperative day 406, mild anemia was detected on hematologic examination, and the seal appeared to have begun losing body condition around this time. On postoperative day 485, decreased food intake was noted, and buprenorphine (0.9–1.2 mg IM) was administered intermittently BID to maintain appetite, with the dose adjusted according to the degree of food intake based on an estimated body weight of 80–100 kg. As emaciation became more pronounced,

prednisolone (60 mg IM BID, Prednisolone Injection; Riken Vets Pharma Co., Ltd., Kawaguchi, Japan), based on an estimated body weight of 60 kg, was additionally administered. Despite continued feeding, the seal gradually became cachectic, and by postoperative day 550, it appeared markedly debilitated and no longer hauled out onto land. From the following day, the seal ceased feeding, and on postoperative day 552, tachypnea was observed. To prevent drowning, the seal was transferred to a holding area without a pool, and flow-by oxygen therapy, prednisolone (60 mg IM), and subcutaneous fluid therapy with 500 mL of physiological saline were administered. Despite these interventions, the seal died on postoperative day 553.

A necropsy was performed on the day of death, and the body weight was 52 kg. Gross and histopathological examination revealed locally recurrent SCC in the mandible, characterized by invasive proliferation into the muscular layer and mandibular bone. The mandibular bone surrounded by the tumor showed osteolysis and necrosis. In addition, a 10.6-mm metastatic lesion was identified in the left mandibular lymph node (Fig. 5), and histopathological examination revealed proliferation of neoplastic squamous epithelial cells similar to those observed in the mandibular mass. No other tumor-related gross lesions were identified. In the right gluteal region, brownish discoloration of the muscle was observed at a site of preoperative buprenorphine administration. Histopathological examination of this area revealed multifocal to coalescing inflammatory changes extending from the subcutaneous tissue to the muscular layer. Gross and histopathological findings unrelated to the tumor included congestive

137 pulmonary edema in the lungs. In the liver, multifocal yellowish-white lesions were observed,
138 characterized by focal necrosis, centrilobular hemosiderin deposition in hepatocytes, lymphocytic and
139 plasmacytic infiltration within portal triads, and a hepatocellular adenoma. In the kidney, mild multifocal
140 lymphocytic infiltration and fibrosis were observed in the renal interstitium. In the aorta, fibrous
141 thickening of the aortic intima with cholesterol deposition was observed. In the uterus, vascular wall
142 calcification was observed in the endometrium and myometrium of both uterine horns. In the spinal cord,
143 mild meningitis predominantly affecting the cervical spinal cord with calcified deposits in the meninges
144 (psammoma bodies), was observed. In both eyes, corneal reddening and opacity were observed, and
145 histopathological examination revealed chronic keratitis and lens degeneration.

146 SCC is one of the most common types of oral tumors in dogs and cats and is characterized by
147 locally invasive behavior [1, 13]. Sites of occurrence are diverse, including the lips, tongue, gingiva,
148 maxilla, mandible, pharynx, and tonsillar region. The tumor is often ulcerative and may exhibit bone
149 proliferation or osteolysis. Clinical signs typically include decreased appetite, anorexia, lethargy, weight
150 loss, ptyalism, and halitosis [1]. Surgical excision remains the mainstay of treatment.

151 In pinnipeds, SCC has also been reported to exhibit aggressive biological behavior. In cases of
152 esophageal SCC, metastatic lesions have been identified in regional lymph nodes, the pancreas, liver,
153 pancreatic lymph nodes, hilar lymph nodes, and periaortic lymph nodes [4, 10]. In addition, pancreatic
154 SCC has been reported to metastasize to the lungs, liver, spleen, mesentery, and mediastinal lymph nodes

[12]. Furthermore, in some pinniped cases, SCC has been histologically confirmed in multiple organs, including the lungs, stomach, omentum, kidneys, reproductive organs, and lymph nodes at necropsy, making identification of the primary lesion difficult [6, 7]. Oral SCC involving the mandibular gingiva or lingual has been reported in California sea lions (*Zalophus californianus*). In the gingival case, no metastatic lesions were identified, whereas in the lingual case, metastasis to the mandibular lymph node was reported [2, 11]. Affected animals generally developed anorexia and had a poor prognosis, often dying or being euthanized before active treatment could be attempted [1, 2, 11]. To the best of our knowledge, this report is the first to describe mandibulectomy in a pinniped.

In domestic animals such as dogs and cats, partial mandibulectomy is generally well tolerated without major impact on feeding function; however, bilateral mandibulectomy extending caudal to the second premolar, representing removal of approximately 50% or more of the mandible, is known to cause feeding difficulty [3, 5, 9].

In addition, a minimum surgical margin of 2 cm is recommended for oral SCC in dogs and cats [5, 9]. In the present case, achieving a margin of ≥ 2 cm would have required the removal of more than half of the mandible, raising concerns regarding loss of feeding ability.

Therefore, marginal resection with minimal margins was selected to achieve both pain relief and preservation of oral function. As a result, the resection extended from the right second premolar to the left second incisor, involving loss of seven of 16 mandibular teeth (approximately 44%). This extent

of resection remained below the 50% threshold reported to cause feeding impairment in dogs and cats, suggesting functional feasibility.

Furthermore, achieving a 2-cm margin would have inevitably resulted in lip defects. By preserving the lips and reconstructing their morphology, the structural integrity of the anterior oral region required for food capture was maintained. Consequently, the seal could grasp fish effectively, not only taking fish from caretakers but also catching and consuming fish independently underwater. These observations highlight the importance of lip preservation and reconstruction for postoperative feeding function (Fig. 3). During mandibulectomy, partial resection of the corresponding lip may allow a more physiological reconstructive outcome without redundant tissue and may reduce the risk of local recurrence. However, in the present case, the lips were preserved (Fig. 3). This decision was based on concern that excision of redundant lip tissue could shorten the mandible relative to the maxilla, potentially resulting in a persistent open-mouth posture. Because seals are aquatic animals that swim and dive, inadequate oral closure was considered likely to allow water to enter the oral cavity. In this case, postoperative swimming and feeding were maintained, suggesting that lip preservation was functionally beneficial. In addition, seals are considered to have less lip laxity than dogs and cats; therefore, simultaneous resection of the jaw and lip may increase the risk of wound dehiscence due to excessive tension, indicating that careful consideration is warranted.

190 Feeding resumed within 2 days postoperatively, and even after tumor recurrence, the animal
191 maintained the ability to eat, suggesting that osteolysis-associated cancer pain had been alleviated. Bone
192 invasion–related pain is known to be severe and is driven by tumor-induced osteoclast activation and
193 subsequent production of inflammatory mediators, cytokines, and prostaglandins, leading to cancer-
194 associated bone pain, functional impairment, and reduced quality of life (QOL) [14].

195 In the present case, removal of the affected bone likely prevented recurrence of bone pain after
196 tumor relapse, allowing continued feeding and maintenance of good QOL.

197 Regarding the cause of death, although neoplastic lesions were identified in the mandible and
198 lymph nodes, the animal remained capable of feeding until shortly before death, suggesting that local
199 tumor effects were unlikely to have been fatal. In addition, considering the size of the tumor and the
200 clinical course, it was considered unlikely that the neoplastic disease exerted sufficient systemic effects
201 to induce cancer-associated cachexia. Therefore, the neoplastic lesions were considered unlikely to have
202 been directly related to the cause of death.

203 In contrast, histopathological examination revealed pulmonary edema and hepatic necrosis,
204 findings suggestive of circulatory failure accompanied by congestion; however, no apparent organic
205 disease capable of explaining the circulatory failure was identified. Taken together with the animal's
206 advanced age, these findings suggest that an age-related decline in circulatory function may have
207 contributed to death; however, the precise cause of death could not be definitively determined.

208 In summary, the feeding disorder in this case was attributed to cancer-associated pain caused
209 by osteolytic invasion by oral SCC. Partial mandibulectomy effectively relieved pain, allowing the
210 animal to eat for more than 1 year postoperatively with preserved QOL. This case demonstrates that
211 partial mandibulectomy can be safely performed in seals and that resection limited to the region between
212 the right second premolar and left second incisor can preserve feeding function.

213 **CONFLICT OF INTEREST**

214 The authors declare no potential conflicts of interest with respect to the research, authorship, or
215 publication of this article.

216

217 **ACKNOWLEDGEMENTS**

218 The authors thank the staff of Otaru Aquarium for their dedicated care and management of the animal
219 described in this report.

REFERENCES

- [1] Bilgic O, Duda L, Sanchez MD, Lewis JR. 2015. Feline oral squamous cell carcinoma: Clinical manifestations and literature review. *J Vet Dent* **32**: 30–40.
- [2] Bossart GD. 1990. Invasive gingival squamous cell carcinoma in a California sea lion (*Zalophus californianus*). *J Zoo Wildl Med* **21**: 92–94.
- [3] Cray M, Selmic LE, Kindra C, Abrams B, Story A, Hovis K, Wustefeld-Janssens B, Park K, Grimes JA, Oblak M, Chen C, Casale S, Rollins A, Kennedy K, Wilson J, Culp WTN, Wavreille VA. 2021. Analysis of risk factors associated with complications following mandibulectomy and maxillectomy in dogs. *J Am Vet Med Assoc* **259**: 265–274.
- [4] Flower JE, Gamble KC, Stone M, Lyons JA, Maganti RJ, Tuomi PA, Olds JE, Sims MA, Gauger P, Tuttle AD. 2014. Esophageal squamous cell carcinoma in six harbor seals (*Phoca vitulina* spp.). *J. Zoo Wildl Med* **45**: 620–631.
- [5] Griffon D, Hamaide A. 2016. *Complications in Small Animal Surgery*, 1st ed., Wiley-Blackwell, Ames, IA, USA.
- [6] Joseph BE, Cornell LH, Migaki G. 1986. Metastatic squamous cell carcinoma in a beached California sea lion (*Zalophus californianus*). *J Wildl Dis* **22**: 281–283.

- 236 [7] Michishita M, Torikai K, Yoshimura H, Terasawa F, Nakahira R, Ohkusu-Tsukada K, Takahashi K.
237 2013. Squamous cell carcinoma with systemic metastases in a spotted seal (*Phoca largha*). *Jpn J*
238 *Zoo Wildl Med* **18**: 129–132.
- 239 [8] Newman SJ, Smith SA. 2006. Marine mammal neoplasia: a review. *Vet Pathol* **43**: 865–880.
- 240 [9] Northrup NC, Selting KA, Rassnick KM, Kristal O, O’Brien MG, Dank G, Dhaliwal RS, Jagannatha
241 S, Cornell KK, Gieger TL. 2006. Outcomes of cats with oral tumors treated with mandibulectomy:
242 42 cases. *J Am Anim Hosp Assoc* **42**: 350–360.
- 243 [10] Ohno K, Fukui D. 2020. Esophageal squamous cell carcinoma in a spotted seal (*Phoca largha*)
244 upon examination by contrast-enhanced computed tomography. *Jpn J Zoo Wildl Med* **26**:11–16.
- 245 [11] Sato T, Higuchi T, Shibuya H, Ohba S, Nogami S, Shirai W, Watanabe H, Honda S. 2002.
246 Lingual squamous cell carcinoma in a California sea lion (*Zalophus californianus*). *J Zoo Wildl*
247 *Med* **33**: 367–370.
- 248 [12] Tsuka T, Kozu T, Sunden Y, Morita T, Okamoto Y, Yamashita M, Osaki T, Amaha T, Ito N, Murahata
249 Y, Imagawa T. 2022. Detection of squamous cell carcinoma of presumed pancreatic origin and its
250 metastasis in a spotted seal (*Phoca largha*) using ultrasonography and computed tomography. *J Vet*
251 *Med Sci* **84**: 373–377.
- 252 [13] Vos JH, van der Gaag I. 1987. Canine and feline oral-pharyngeal tumours, *Zentralbl Veterinarmed*
253 *A* **34**: 420–427.

254 [14] Zajackowska R, Kocot-Kepska M, Leppert W, Wordliczek J. 2019. Bone pain in cancer patients:
255 mechanisms and current treatment. *Int J Mol Sci* **20**: 6047.
256

FIGURE LEGENDS

Fig. 1. Macroscopic and computed tomographic (CT) images of a tumor in the rostral mandible of a spotted seal. **a.** Ulceration is present on the lesion surface, and the two right mandibular incisors are not visible due to the mass. **b.** Contrast-enhanced CT revealed a $25 \times 30 \times 45$ mm mass with indistinct margins, osteolysis, and bone proliferation in the region of the right mandibular canine root, accompanied by mandibular enlargement. **c.** Three-dimensional volume-rendered CT image showing an osteolytic mass involving the rostral mandible, with marked enlargement and deformation.

Fig. 2. Intraoperative photographs. **a.** The gingival and sublingual mucosa were incised, and the mandibular body was exposed using monopolar electrocautery. **b.** A high-speed surgical drill with a round bur was used to approach the mandibular body laterally between the right second and third premolars. **c.** Hemorrhage from the inferior alveolar artery and vein. **d.** The rostral mandible was completely resected.

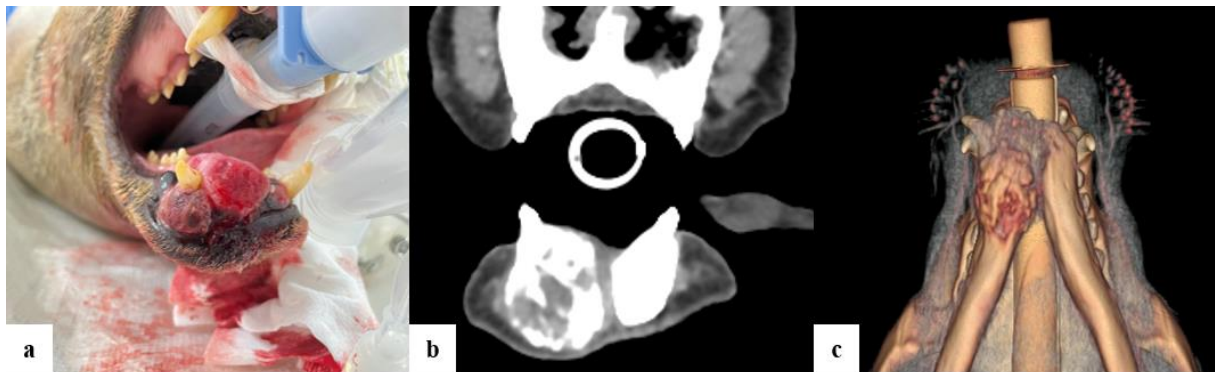
Fig. 3. Photographs of lip reconstruction. **a.** Right oblique intraoperative view during reconstruction. **b.** Frontal view of the rostral mandible after partial mandibulectomy and reconstruction.

Fig. 4. Low- and high-power photomicrographs of the surgically excised mandibular mass in a spotted seal. **a.** Low-power view showing island-like invasive proliferation of keratinizing neoplastic epithelial cells within mandibular tissue (H&E, $\times 40$; bar = 200 μm). **b.** High-power view demonstrating marked nuclear pleomorphism and mitotic figures (5 per high-power field). Reactive proliferation of fibroblastic stromal cells and neutrophil infiltration are also present.

Fig. 5. Cut surface of the left mandibular lymph node showing a white lesion (white arrowheads) observed at necropsy. Histopathological examination revealed proliferation of neoplastic squamous epithelial cells similar to those observed in the mandibular mass, confirming metastasis.

284 **FIGURE**

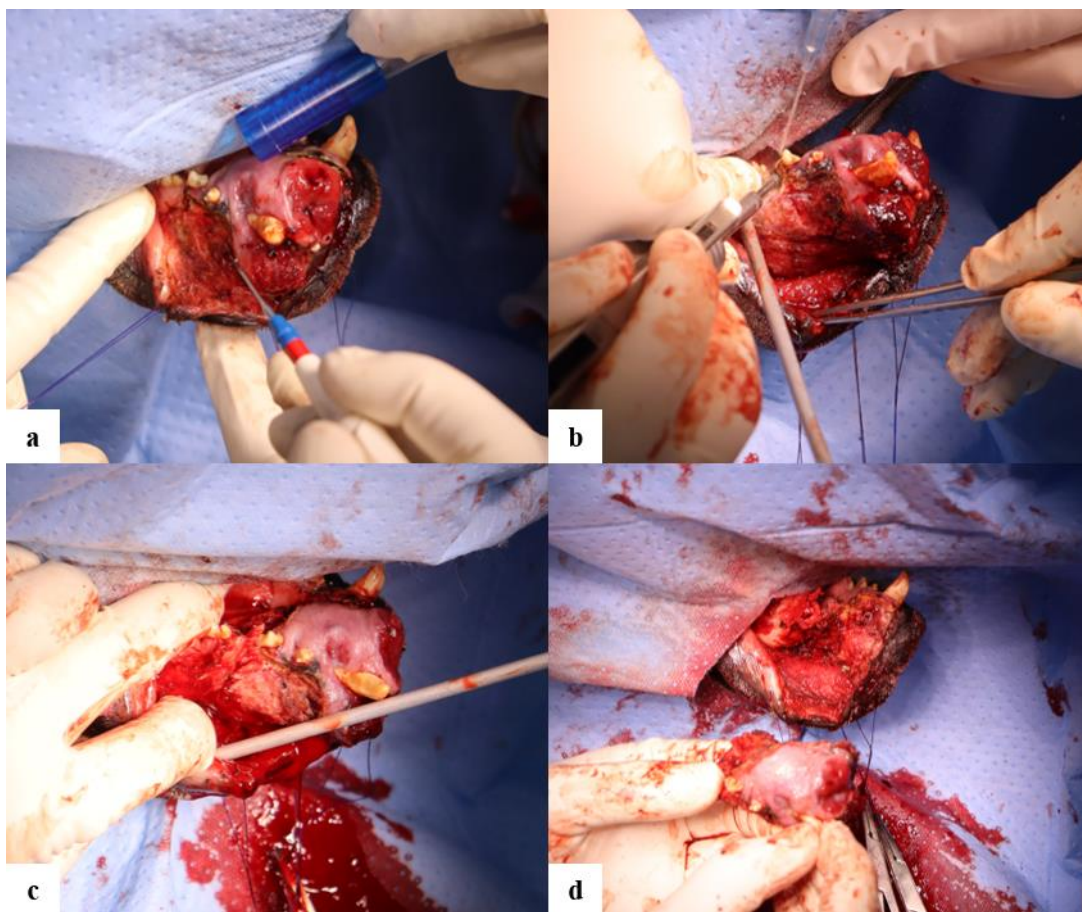
285



286

287 **Fig. 1**

288

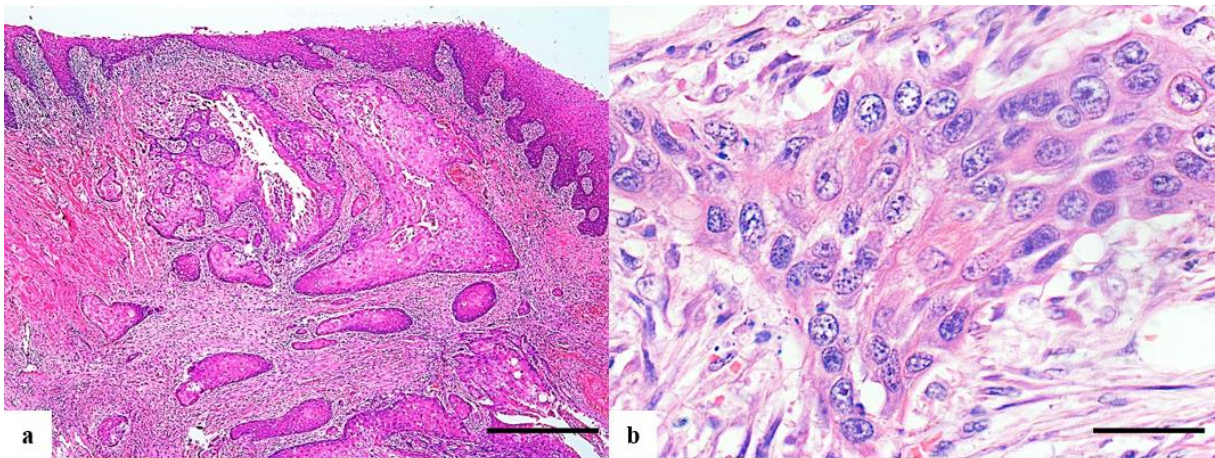


289

290 **Fig. 2**



292 **Fig. 3**



295 **Fig. 4**



297

298 **Fig. 5**